

QCar2 Internship

Quang Thinh BUI

May 2026

QCar2: Non-Linear MPC for trajectory tracking

““latex

1 Introduction

For trajectory planning, the QCar2 is approximated by a kinematic single-track model, which provides a simpler representation than the full dynamic model. Let X and Y denote the Cartesian coordinates of the vehicle in the inertial frame, ψ the vehicle heading angle, and v_x the longitudinal velocity. Let δ be the front-wheel steering angle, a_x the longitudinal acceleration, and l the wheelbase of the vehicle. Moreover, the Cartesian coordinates of the midpoint of the front axle are given by

$$x_f = x + l \cos \psi, \quad y_f = y + l \sin \psi. \quad (1)$$

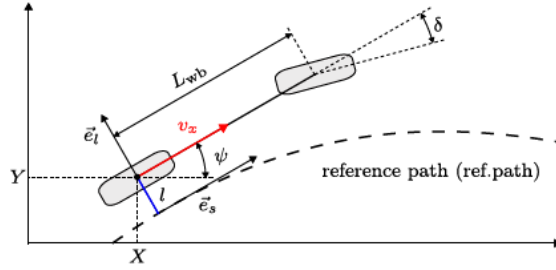


Figure 1: Car-like vehicle model.

The state and input vectors are defined as

$$\mathbf{q}_k = [X \quad Y \quad \psi \quad v_x]^\top, \quad \mathbf{u}_k = [\delta \quad a_x]^\top.$$

Under the pure-rolling and no-slip assumptions, the kinematic single-track model is given by

$$\dot{X} = v_x \cos \psi, \quad (2a)$$

$$\dot{Y} = v_x \sin \psi, \quad (2b)$$

$$\dot{\psi} = \frac{v_x}{l} \tan \delta, \quad (2c)$$

$$\dot{v}_x = a_x. \quad (2d)$$

2 Nonlinear Model Predictive Control

In this section, the trajectory tracking problem is formulated as a nonlinear model predictive control problem over a finite prediction horizon. The state and input vectors of the QCar2 are defined as

$$\mathbf{q} = [X \quad Y \quad \psi \quad v_x]^\top, \quad \mathbf{u} = [\delta \quad a_x]^\top.$$

The reference trajectory generated by the flatness-based trajectory planner is represented by the reference state and input sequences

$$\mathbf{q}_{\text{ref},i} = [X_{\text{ref},i} \quad Y_{\text{ref},i} \quad \psi_{\text{ref},i} \quad v_{x,\text{ref},i}]^\top, \quad \mathbf{u}_{\text{ref},i} = [\delta_{\text{ref},i} \quad a_{x,\text{ref},i}]^\top.$$

The admissible state set is defined by the longitudinal velocity constraint:

$$\mathcal{Q} = \{\mathbf{x} \in \mathbb{R}^4 : v_{x,\min} \leq v_x \leq v_{x,\max}\}.$$

The admissible input set is defined by the steering angle and longitudinal acceleration constraints:

$$\mathcal{U} = \{\mathbf{u} \in \mathbb{R}^2 : \delta_{\min} \leq \delta \leq \delta_{\max}, \quad a_{x,\min} \leq a_x \leq a_{x,\max}\}.$$

Given a prediction horizon $N > 0$, the nonlinear optimal control problem solved at each sampling time is written as

$$\begin{aligned} \min_{\mathbf{q}_0, \dots, \mathbf{q}_N, \mathbf{u}_0, \dots, \mathbf{u}_{N-1}} \quad & \sum_{i=0}^{N-1} \left[(\mathbf{q}_i - \mathbf{q}_{\text{ref},i})^\top Q (\mathbf{q}_i - \mathbf{q}_{\text{ref},i}) + (\mathbf{u}_i - \mathbf{u}_{\text{ref},i})^\top R (\mathbf{u}_i - \mathbf{u}_{\text{ref},i}) \right] \\ & + (\mathbf{q}_N - \mathbf{q}_{\text{ref},N})^\top P (\mathbf{q}_N - \mathbf{q}_{\text{ref},N}) \\ \text{s.t.} \quad & \mathbf{q}_0 = \mathbf{q}_{\text{meas}}, \\ & \mathbf{q}_{i+1} = f_{\text{ERK}}(\mathbf{q}_i, \mathbf{u}_i), \quad i = 0, \dots, N-1, \\ & \mathbf{q}_i \in \mathcal{Q}, \quad i = 0, \dots, N, \\ & \mathbf{u}_i \in \mathcal{U}, \quad i = 0, \dots, N-1. \end{aligned}$$

Here, $Q = Q^\top \geq 0$, $R = R^\top > 0$, and $P = P^\top \geq 0$ are the weighting matrices for the state tracking error, input tracking error, and terminal state tracking error, respectively.

In the implementation, the continuous-time kinematic single-track model is provided to acados and discretized internally using an explicit Runge-Kutta integrator. Therefore, f_{ERK} denotes the discrete-time prediction model obtained from the continuous-time vehicle dynamics.

At each sampling time, the measured vehicle state is imposed as the initial condition. The reference state and input sequences

$$\mathbf{q}_{\text{ref},0}, \mathbf{q}_{\text{ref},1}, \dots, \mathbf{q}_{\text{ref},N}, \quad \mathbf{u}_{\text{ref},0}, \mathbf{u}_{\text{ref},1}, \dots, \mathbf{u}_{\text{ref},N-1}$$

are then updated from the generated trajectory. The nonlinear optimal control problem is solved using acados, and only the first optimal input is applied to the vehicle:

$$\mathbf{u}_{\text{applied}} = \mathbf{u}_0^*.$$

At the next sampling time, the vehicle state is measured again and the same optimization procedure is repeated in a receding horizon fashion. ““